HARRISON MUHORO

FULL-STACK SOFTWARE ENGINEER • LARAVEL • REACT • TYPESCRIPT • NODE.JS

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer and Diploma candidate specialized in modernizing paper-heavy operational workflows into secure, high-throughput digital systems. Proven track record of building production-grade enterprise software—including multi-tenant hospital portals, hotel management systems, and government logistics tools. Deep focus on reliable backend architecture: designing strict tenant isolation, real-time WebSockets, bulletproof audit logging, and local payment integration (M-Pesa Daraja). Seeking a full-stack or backend engineering role focused on scaling robust systems.

TECHNICAL SKILLS

Languages & Frameworks: PHP 8.3, Laravel 11, TypeScript, JavaScript (Node.js/Express), React 18/19, Blade, HTML5/CSS3, Tailwind CSS

Databases & Caching: MySQL 8, PostgreSQL, Redis, Prisma ORM

Architecture & Security: Role-Based Access Control (RBAC), Multi-Tenancy, Global Scoping, Audit Logging, Rate Limiting, JWT Auth, REST APIs

Real-Time & Integrations: Laravel Reverb (WebSockets), M-Pesa Daraja (STK Push & C2B), Africa's Talking SMS, Resend API

Testing & DevOps: PHPUnit, Pest, Vite, Git, Docker, Nginx, CI/CD, Azure (AZ-900 in progress)

SELECTED PROJECTS

Afya HMS — Multi-Branch Hospital Management System

github.com/harrisonmuhoro/afyahms

`Laravel 11 • React 18 • TypeScript • MySQL 8 • Redis • Laravel Reverb • M-Pesa PWA`

- Engineered a modular monolith managing end-to-end workflows across front desk, clinic, pharmacy, lab, and admin operations.
- Implemented real-time patient queue updates via Laravel Reverb (WebSockets) to eliminate waiting room bottlenecks.
- Enforced strict branch-level tenant isolation using global query scopes, independently identifying and fixing a critical authorization bypass vulnerability on the `Patient` model.
- Integrated M-Pesa Daraja for automated point-of-sale payment validation and reconciliation.

Taara HMS — Hotel Management & POS System

github.com/harrisonmuhoro/Taara-HMS

`Laravel 11 • Blade • TypeScript • Vite • MySQL 8 • M-Pesa Daraja`

- Architected a lightweight, server-rendered application tailored for low-spec legacy POS terminals across front desk, housekeeping, and restaurant operations.
- Built automated payment matching via M-Pesa STK Push and C2B webhooks alongside queued async background job processing for PDF/CSV financial reporting.
- Designed comprehensive audit log tracking for all financial transactions and implemented per-IP/account rate limiting with automated lockouts and admin overrides.

TuaVois ERP — Multi-Tenant Travel Agency Operations Engine

github.com/harrisonmuhoro/Tuavois

`Node.js • Express 5 • TypeScript • Prisma • MySQL 8 • React 19 • TanStack Query • Zod`

- Developed a multi-tenant backend handling CRM, automated quote generation, invoicing, and trip management under 6 role levels.
- Leveraged serializable database transactions to eliminate race conditions during concurrent financial payments, backed by 14 integration test suites on an isolated test database.

InduTrack KE — Industrial Attachment Management Platform

github.com/harrisonmuhoro/InduTrack-Ke

`Laravel 11 • React • TypeScript • Vite • MySQL • Sanctum • Africa's Talking`

- Digitized paper-based industrial logbooks into a tamper-proof tracking platform supporting six distinct user tiers.
- Integrated Africa's Talking SMS API for instant student/supervisor notifications and automated background PDF generation for official compliance certificates.

Vuka Attachment Portal — Government Internship Desk

github.com/harrisonmuhoro/vuka_attachment_portal

`PHP • MySQL • JavaScript • Chart.js`

- Developed an internship management portal featuring email-verified authentication and a four-tier RBAC system with strict department isolation.
- Built interactive data visualizations using Chart.js and implemented reliable CSV export functionality for administrative reporting.

WORK EXPERIENCE

IT Support Attachment Student — **Department of ICT, Nyeri County Government** Sep 2025 – Nov 2025 & Sep 2026 – Nov 2026

- Delivered level-1 and level-2 IT support across multiple government offices, maintaining network uptime, workstation hardware, and operating systems.
- Provisioned, configured, and deployed workstations and network peripherals, drastically reducing diagnostic turnaround times for staff hardware and software issues.

EDUCATION

Diploma in Software Development
Nyeri National Polytechnic • 2024 – Expected Mar 2028

KCSE (Mean Grade: B-)
Mayori Secondary School • 2020 – 2023

CERTIFICATIONS

- **Data Analytics Training** – ICT Authority, Kenya (Jul 2026)
- **Back End Development & APIs** – freeCodeCamp (2024)
- **Responsive Web Design** – freeCodeCamp (2023)
- **Microsoft Azure Fundamentals (AZ-900)** – In Progress

COMMUNITY & LANGUAGES

Audio-Visual Technician: Deliverance Church International Skuta (Live sound, projection, and production) | **Languages:** English (Professional), Kiswahili (Native)